ESP32

C1
4.7u
C2
0.1u
3V3_ESP32
U3
ESP32-C6-WROOM-1-N8
EN_ESP32
JTAG_TMS_DAP_SWDIO
JTAG_TDI
JTAG_TCK_DAP_SWCLK
JTAG_TDO
FLASH_CS
WIZ_CS
DISP_D/C
ESP_NEOPIXEL
BUZZER_PWM
USB_D-
USB_D+
R39 10K
22R R3
22R R4
MOD1_CS
MOD2_CS
TX_ESP32
RX_ESP32
GPIO15
LCD_CS
I2C_SCL
I2C_SDA
SPI_CLK
SPI_MISO
SPI_MOSI
BOOT_ESP32
22R R8
22R R9
22R R10

STM32

U7
STM32F103C8T6
3V3_STM32
VBAT
PC13-TAMPER-RTC
PC14-OSC32_IN
PC15-OSC32_OUT
PD0-OSC_IN
PD1-OSC_OUT
NRST
VSSA
VDDA
PA0-WKUP
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB10
PB11
VSS_1
VDD_1
VDD_3
VSS_3
PB9
PB8
BOOT0
PB7
PB6
PB5
PB4
PB3
PA15
PA14
VDD_2
VSS_2
PA13
PA12
PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
STM32_TARGET_VCC
SCA_TRIG_OUT
FI_TRIG_IN
SCA_IO1
SCA_IO2
SCA_IO3
STM32_NEOPIXEL
R15 10K
SWDIO
PA12
PA11
STM32_RX
STM32_TX
SCA_CLK
PB15
PB14
PB13
PB12

X1
HCI_0132M4_8MHz
C12
22pF
C10
22pF
HSE_IN
HSE_OUT

STM32_TARGET_VCC
C13
100nF
C14
100nF
C15
100nF
C16
4.7uF
C17
1uF
C18
10nF

BOOT & RESET STM32

SW1
BOOT
3V3_STM32
R25
10K
BOOT_STM32
C19
0.1u
SW2
RESET
R27
10K
NRST
C20
100nF

EEPROM

U2
M24256E-FMN6TP
R5
10K
100nF
C3
3V3_ESP32
EEPROM_WP
I2C_SCL
I2C_SDA
R2
4.7K
R6
4.7K

FLASH

U5
W25Q64JVSSIQ
3V3_ESP32
R13
10K
FLASH_CS
SPI_MISO
R11
10K
100nF
C7
SPI_CLK
SPI_MOSI
R17
10K

disp
File: untitled.kicad_sch

H1
M4_X6521WVS-2x02H-C60D74
STM32_TX
STM32_RX
RX_ESP32
TX_ESP32
H3
M4_X6521WVS-2x02H-C60D74
SWDIO
NRST
SWCLK
H2
210SMT-2*3P Gold-plated black
JTAG_TDI
JTAG_TMS_DAP_SWDIO
JTAG_TDO
JTAG_TCK_DAP_SWCLK

BOOT & RESET ESP32

SW4
BOOT
3V3_ESP32
R26
10K
BOOT_ESP32
C22
0.1u
SW3
RESET
R24
10K
EN_ESP32
C21
1u

TX & RX ESP32

LEDRX1
LEDTX1
LTST-C170KRKT
LTST-C170KBKT
R12
5.1k
R16
5.1k
RX_ESP32
TX_ESP32

IO connectors

U10
F1_PM2.54-1x7PLT-H8.5-R
U11
F1_PM2.54-1x7PLT-H8.5-R
U13
F1_PM2.54-1x7PLT-H8.5-R
U14
F1_PM2.54-1x7PLT-H8.5-R
3V3_ESP32
EN_ESP32
JTAG_TMS_DAP_SWDIO
JTAG_TDI
JTAG_TCK_DAP_SWCLK
JTAG_TDO
FLASH_CS
WIZ_CS
DISP_D/C
ESP_NEOPIXEL
BUZZER_PWM
USB_D-
USB_D+
MOD1_CS
MOD2_CS
TX_ESP32
RX_ESP32
GPIO15
LCD_CS
I2C_SCL
I2C_SDA
SPI_CLK
SPI_MISO
SPI_MOSI
BOOT_ESP32

J1
TYPE-C-31-M-12
VUSB
R1
5.1K
R7
5.1K
PWR_FLAG
C4
10uf

U1
USBLC6-2SC6
C5
100nF

3v3 LDO ESP32

U6
AP2114H-3.3TRG1
VUSB
3V3_ESP32
VIN
C6
100uF
C8
100uF

3v3 LDO STM32

U12
AP2114H-3.3TRG1
3V3_STM32
C9
100uF
C11
100uF

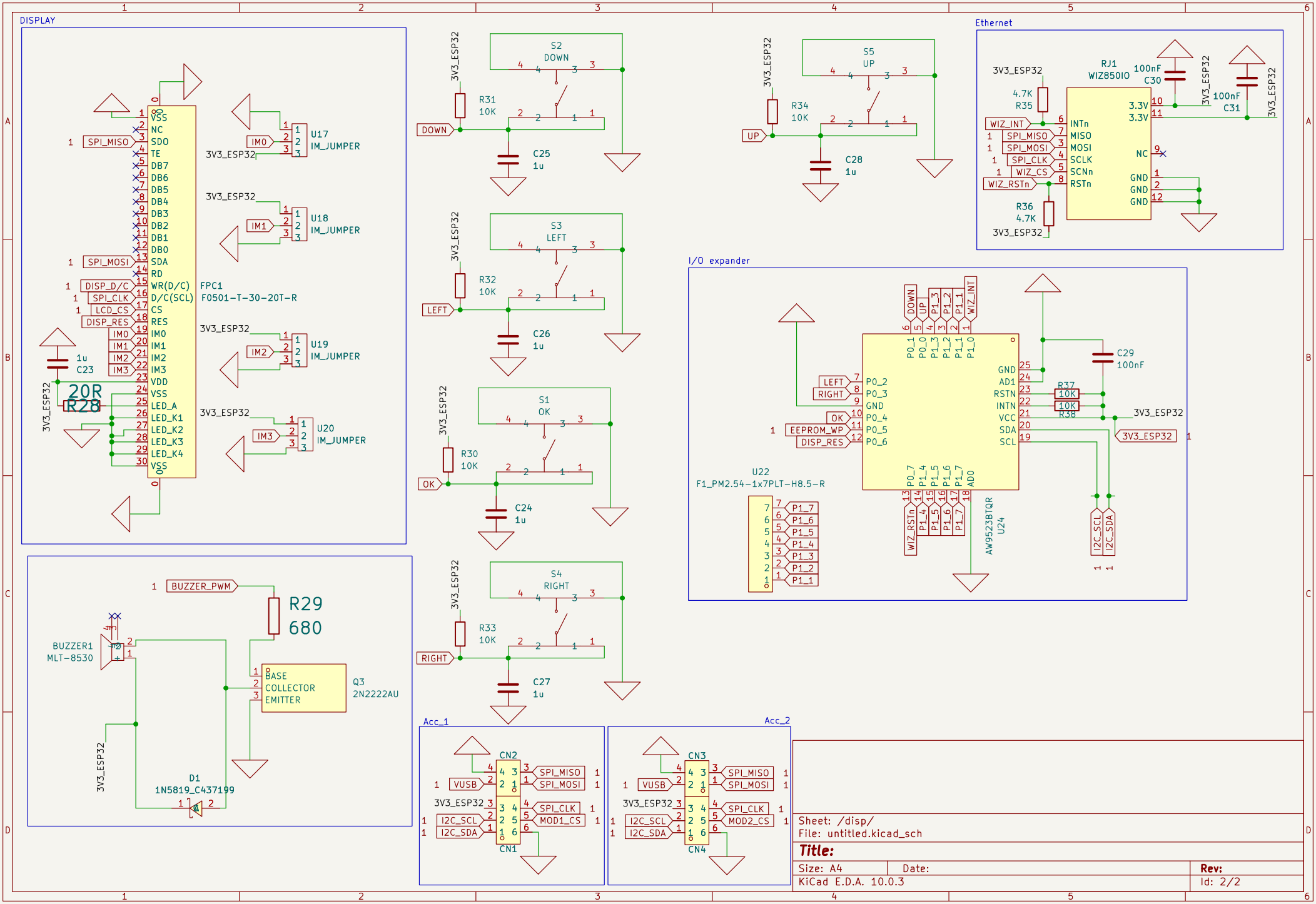
SCA

U15
F1_PM2.54-1x7PLT-H8.5-R
U27
F2_PM2.54-1x5PLT-H5.0-R
SCA_TRIG_OUT
FI_TRIG_IN
SCA_CLK
SCA_IO1
SCA_IO2
SCA_IO3
STM32_TARGET_VCC
R_SHUNT1
0.1R

H5
M3_X6511WVS-03H-C60D48R2
HSE_IN
HSE_OUT

U4
F1_PM2.54-1x7PLT-H8.5-R
U8
F1_PM2.54-1x7PLT-H8.5-R
U9
F1_PM2.54-1x7PLT-H8.5-R
U25
F1_PM2.54-1x7PLT-H8.5-R
PA15
PA12
PA11
PA7
PA6
PA5
PB15
PB14
PB13
PB12
PB11
PB10
PB9
PB8
PB7
PB6
PB5
PB4
PB3
PC15
PC14
PC13

Sheet: /
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Size: A3
Date:
Rev:
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Id: 1/2


DISPLAY
FPC1
F0501-T-30-20T-R
U17
IM_JUMPER
U18
IM_JUMPER
U19
IM_JUMPER
U20
IM_JUMPER
20R
R28
1u
C23
S2
DOWN
R31
10K
C25
1u
S3
LEFT
R32
10K
C26
1u
S1
OK
R30
10K
C24
1u
S4
RIGHT
R33
10K
C27
1u
S5
UP
R34
10K
C28
1u
Ethernet
RJ1
WIZ850IO
100nF
C30
100nF
C31
4.7K
R35
4.7K
R36
I/O expander
U22
F1_PM2.54-1x7PLT-H8.5-R
AW9523BTQR
U24
C29
100nF
R37
10K
R38
10K
BUZZER1
MLT-8530
R29
680
Q3
2N2222AU
D1
1N5819_C437199
Acc_1
Acc_2
CN1
CN2
CN3
CN4
3V3_ESP32
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 2/2